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Mini Review

Gene Therapy for Rare and Neurodegenerative Diseases

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Running title: A concise synthesis of clinical and regulatory progress, current as of September 2026

Abstract

Gene therapy has progressed from an experimental concept to a diversifying clinical toolkit spanning AAV-based in vivo gene delivery, ex vivo cell-based editing, antisense oligonucleotides, and lipid-nanoparticle-delivered CRISPR/base editing. Several dozen cellular and gene therapy products are now licensed by the FDA, and many more are in clinical development, with approvals concentrated in hematologic, neuromuscular, metabolic, and CNS disorders. In rare monogenic disease, 2025 to 2026 brought major milestones: FDA accelerated approval of Genglycos for glycogen storage disease type Ia, published comparative data on the two FDA-approved sickle cell gene therapies (exa-cel, lovo-cel), and, most notably, the first patient-specific in vivo CRISPR base-editing therapy, designed and manufactured within about six months for an infant with CPS1 deficiency. That case was used by FDA leaders to illustrate the plausible mechanism approach, which the agency set out in February 2026 as draft, non-binding guidance (the Plausible Mechanism Framework) describing how individualized therapies might be evaluated using mechanistic, natural-history, and biomarker evidence when randomized trials are not feasible. In neurodegenerative disease, uniQure's AMT-130 reported approximately 75% slowing of Huntington's disease progression at 36 months in the high-dose group compared with a propensity score-matched external control (company-reported topline data; a Biologics License Application was submitted in September 2026), while Parkinson's programs (AAV-GDNF, AAV-GAD) have shown feasibility and safety, with early efficacy signals that still require confirmation in larger controlled trials. Tofersen, an antisense oligonucleotide for SOD1-ALS, is now being tested in the phase 3 ATLAS trial as a template for presymptomatic gene-targeted intervention. Persistent challenges include pre-existing anti-AAV antibodies (neutralizing-antibody seroprevalence of roughly 3 to 59% depending on serotype), one-time-dosing constraints, CNS delivery limitations, manufacturing complexity, and long-term safety follow-up of up to 15 years. Overall, the field is moving toward faster, more individualized regulatory approaches while still working through fundamental delivery and durability problems.

Keywords: Gene therapy; rare genetic diseases; neurodegenerative diseases; CRISPR gene editing; AAV vectors; individualized therapies; Huntington's disease; Parkinson's disease; amyotrophic lateral sclerosis; regulatory framework

1. Introduction

Gene therapy has moved from a largely experimental modality to a diversifying clinical toolkit encompassing in vivo viral vector delivery (chiefly adeno-associated virus, AAV), ex vivo cell-based gene modification, antisense oligonucleotides (ASOs), and, most recently, in vivo CRISPR/base editing delivered via lipid nanoparticles (LNPs). Several dozen cellular and gene therapy products are now licensed by the FDA [1], with approvals concentrated in hematologic disorders, neuromuscular disease, and inherited metabolic and CNS conditions, and many more are in clinical development. This review summarizes the most consequential advances of the past 12 to 18 months across rare monogenic diseases and neurodegenerative disorders, together with the earlier approvals

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that frame them, and the regulatory shifts accompanying them.

2. Rare Monogenic Diseases

2.1 Metabolic and hematologic disease approvals

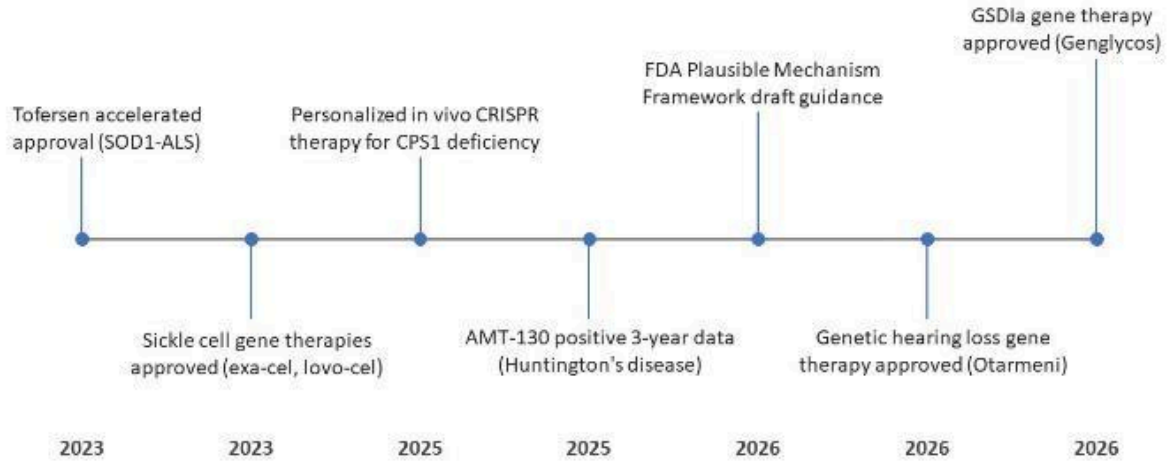


Figure 1: *Timeline of key gene therapy milestones, 2023-2026.*

In August 2026, the FDA granted accelerated approval to Genglycos (pariglasgene brecaparvovec-opnr), an AAV8 gene therapy and the first approved treatment for glycogen storage disease type Ia in patients aged 8 years and older, indicated to reduce daily cornstarch intake as an adjunct to nutritional management [2]. Approval rests on a surrogate endpoint, and the sponsor has committed to an expanded post-marketing disease-monitoring program covering 50 commercially treated patients and 20 untreated controls (patients who cannot be treated because of anti-AAV8 antibodies), within a total monitoring period of 10 years [3]. This follows earlier approvals of ex vivo lentiviral therapies such as Libmeldy (marketed in the United States as Lenmeldy and FDA-approved in March 2024) for metachromatic leukodystrophy [4] and the AAV-based Zolgensma for spinal muscular atrophy, both of which established proof-of-concept for CNS-relevant monogenic disease correction [5]. In hemoglobinopathies, the two FDA-approved sickle cell disease gene therapies, lovo-cel (lentiviral, bluebird bio) and exa-cel (CRISPR-Cas9 ex vivo editing, Vertex/CRISPR Therapeutics), have been compared on published trial design, efficacy, and safety, with both showing durable resolution of severe vaso-occlusive events (more than 90%) but differing in trial populations, stem-cell mobilization and apheresis requirements, and manufacturing [6]. A gene therapy for genetic

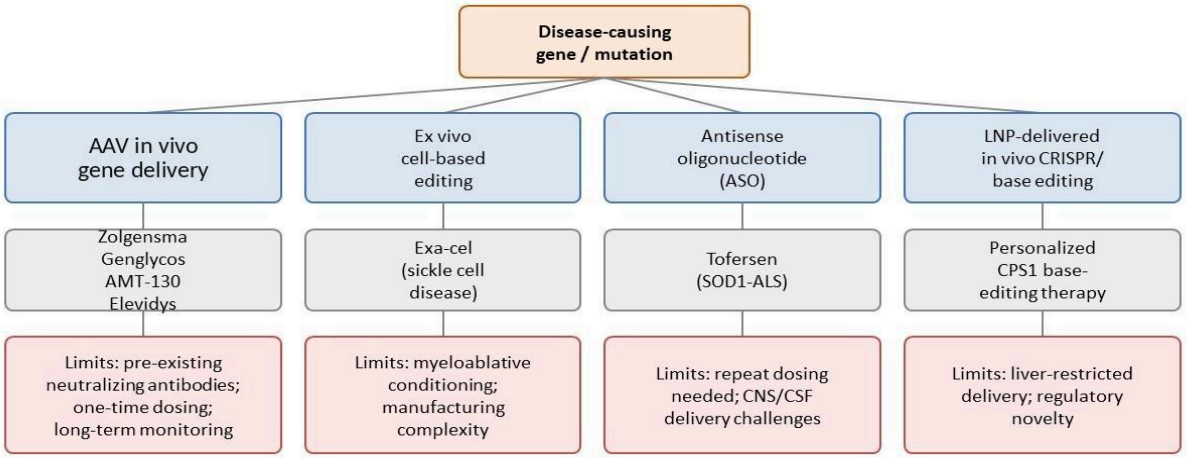


Figure 2: *Major gene-targeted therapeutic platforms in current clinical use, with representative products and key limitations.*

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hearing loss (Regeneron's Otarmeni, a dual-AAV1 therapy for OTOF-associated hearing loss) also received accelerated approval in April 2026 under the FDA's Commissioner's National Priority Voucher pilot [7]. Selected products are summarized in Table 1.

Table 1. Selected Gene Therapies for Rare Monogenic Diseases

Disease	Product	Platform	Status (Sept 2026)	Ref
Glycogen storage disease type Ia	Genglycos (pariglasgene brecaparvovec-opnr)	AAV8, in vivo	FDA accelerated approval, Aug 2026	[2]
Spinal muscular atrophy	Zolgensma (onasemnogene abeparvovec)	AAV9, in vivo	FDA/EMA approved (established)	[5]
Metachromatic leukodystrophy	Libmeldy (US: Lenmeldy)	Lentivirus, ex vivo HSC	EMA approved; FDA approved Mar 2024	[4,5]
Sickle cell disease	Exa-cel (Casgevy)	CRISPR-Cas9, ex vivo HSC	FDA approved, Dec 2023	[6]
Sickle cell disease	Lovo-cel (Lyfgenia)	Lentivirus, ex vivo HSC	FDA approved, Dec 2023	[6]
Genetic hearing loss	Otarmeni	Dual AAV1, in vivo	FDA accelerated approval, 2026	[7]
Severe CPS1 deficiency (single patient)	k-abe (abengcemeran; patient-specific base editor)	LNP-delivered base editor, in vivo	First-in-human (single patient), 2025	[9]
Duchenne muscular dystrophy	Delandistrogene moxeparvovec (Elevidys)	AAVrh74, in vivo	FDA approved (ambulatory patients ≥4 yrs; non-ambulatory indication removed Nov 2025; boxed warning)	[17–19]

2.2 Individualized (N-of-1) gene editing

Perhaps the most conceptually significant development is the first patient-specific in vivo CRISPR base-editing therapy, reported in the New England Journal of Medicine in May 2025. An infant ("KJ") with severe carbamoyl phosphate synthetase 1 (CPS1) deficiency received an LNP-delivered adenine base editor (k-abe) custom-designed to correct his private mutation [8,9]; the bespoke therapy was designed and manufactured within roughly six months, and the first infusion was given in February 2025 at 6 to 7 months of age, with further doses in March and April 2025. By April 2025 he had received three doses with no serious adverse events, tolerated more dietary protein, and needed less nitrogen-scavenger medication, although longer follow-up is needed to establish safety and efficacy [9,10]. Because delivery used LNPs rather than a viral capsid, repeat dosing was possible, an advantage over AAV-based platforms, in which anti-capsid antibodies can limit redosing [9]. Academic and philanthropic efforts are now aiming to scale this approach, including a new Center for Pediatric CRISPR Cures [11]. The case was also used by FDA leaders in a November 2025 New England Journal of Medicine article to illustrate the agency's proposed plausible mechanism approach [12]. In February 2026 the FDA then issued draft, non-binding guidance on a Plausible Mechanism Framework, which describes considerations for generating evidence of safety and effectiveness for individualized genome-editing and RNA-based therapies (and potentially other tailored therapeutics) when randomized trials are not feasible, relying on the identified molecular cause, natural-history data, and confirmation of target editing rather than traditional randomized trials [13–16].

3. Neurodegenerative Diseases

3.1 Huntington's disease

UniQure's AMT-130 (ifezuntirgene inilparvovec), an AAV5 vector delivering a microRNA designed to lower huntingtin, reported encouraging topline data. In results announced in September 2025 (data cutoff 30 June 2025), 12 high-dose patients with 36 months of follow-up showed a 75% slowing of progression on the composite Unified Huntington's Disease Rating Scale (cUHDRS; p=0.003) and a 60% slowing on Total Functional Capacity relative to a propensity score-matched external control drawn from the Enroll-HD natural-history dataset, alongside a mean 8.2% reduction from baseline in CSF

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neurofilament light chain, a marker of neurodegeneration [20–23]. These are company-reported results from a small study without a concurrent control group and should be interpreted cautiously. Regulatory alignment has shifted repeatedly: the FDA indicated in October 2025 and again in March 2026 that comparison with an external control was not sufficient as primary evidence [24], but in June 2026 it indicated that the three-year analysis could serve as the primary basis of an accelerated-approval application [25], and uniQure submitted the Biologics License Application, together with a UK marketing authorization application, on 2 September 2026 [26]. The broader pipeline remains active: Skyhawk's SKY-0515 (an oral mRNA splicing modulator) advanced to a phase II/III trial (FALCON-HD), Spark Therapeutics' SPK-10001 gene therapy entered phase I/II, and PTC518 (votoplam), an oral huntingtin-lowering splicing modulator, met its primary endpoint in the PIVOT-HD study, while Roche's ASO tominersen and Sage's dalzanemdor programs illustrate that not every mechanistic approach has succeeded, underscoring how competitive and mechanistically diverse the HD field has become [27,28].

3.2 Parkinson's disease

Gene therapy for Parkinson's disease is shifting from purely symptomatic dopaminergic/GABAergic modulation toward disease-relevant molecular targets. AB-1005, an AAV2 vector delivering GDNF (glial cell line-derived neurotrophic factor) directly to the putamen, progressed from a phase 1b study, which met its primary safety and efficacy endpoints at 18 months [29], into the randomized, sham-surgery-controlled phase 2 REGENERATE-PD trial [30]. Earlier approaches, AAV2-GAD (increasing inhibitory GABA tone in the subthalamic nucleus) and AAV2-neurturin, established feasibility and acceptable safety in early-phase trials, though efficacy signals have been modest and trial design (particularly sham-surgery control) remains a persistent methodological challenge [31]. More recently, a randomized, sham-controlled phase 1/2 study of AAV-GAD (NCT05603312; 14 participants) was completed in 2024, and the sponsor reported a statistically significant average 18-point improvement in the OFF-medication MDS-UPDRS part III score in the high-dose group at 26 weeks [32,33]; these company-announced findings come from a very small study and await confirmation in larger controlled trials. A 2026 review of the field concludes that clinical trials have now validated durable, neuron-specific transgene expression, but that translating molecular target engagement into consistent clinical benefit is still the central unsolved problem [34].

3.3 Amyotrophic lateral sclerosis (ALS)

Tofersen, an ASO (not a viral gene therapy, but part of the same gene-targeted therapeutic class) directed against SOD1 mRNA, remains the most mature genetically-targeted ALS treatment, having received FDA accelerated approval in 2023 for SOD1-ALS based on reduction of neurofilament light chain [35–37]. SOD1 variants account for a small minority of ALS cases (approximately 2% of sporadic and 11% of familial cases in one European estimate) [38]. In the phase 3 VALOR trial the primary functional endpoint was not met at week 28 in rapidly progressing participants, but reductions in SOD1 and neurofilament light chain and signals of slower decline emerged, particularly with earlier treatment in the open-label extension [39,40]. A systematic review and meta-analysis of 12 studies including 195 tofersen-treated patients reported reductions in CSF SOD1 protein and plasma neurofilament light chain and, in the meta-analysis of randomized trials, a slower decline in ALSFRS-R, although the evidence base was heterogeneous and included observational studies and case reports [38]. The ongoing phase 3 ATLAS trial is testing whether presymptomatic initiation, guided by elevated plasma neurofilament light in carriers of high-penetrance SOD1 variants, can delay or prevent clinical onset altogether, and no efficacy data are yet available [37,41]; the paradigm could serve as a template for presymptomatic gene-targeted intervention in other familial neurodegenerative diseases. Selected programs are summarized in Table 2.

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Table 2. Selected Gene-Targeted Therapy Programs in Neurodegenerative Disease

Disease	Product	Mechanism	Phase / Status	Key Outcome Signal	Ref
Huntington's disease	AMT-130 (uniQure)	AAV5, HTT-lowering microRNA	Ph 1/2; accelerated approval soughtBLA submitted Sept 2026	~75% slowed progression at 3 yrsslowing of cUHDRS decline at 36 mo (high dose vs external control)	[12,13,15,16 21,23,26]
Huntington's disease	SKY-0515 (Skyhawk)	mRNA splicing modulator	Ph II/III (FALCON-HD)	Trial ongoing	[11,27]
Huntington's disease	Votoplam / PTC518	Oral splicing modulator	PIVOT-HD	Met primary endpoint (May 2025)	[10,11,27]
Parkinson's disease	AB-1005	AAV2, GDNF to putamen	Ph 2 (REGENERATE-PD)	Successful putamen coverage (Ph1b)	[1729,30]
Parkinson's disease	AAV2-GAD	AAV2, GABA modulation (STN)	LongPh 1/2 completed (2024); long-term F/U completefollow-up ongoing	Safe; modest efficacyefficacy signal in small study (n=14), unconfirmed	[22,24,25,31 –33]
ALS (SOD1)	Tofersen (Qalsody)	ASO, SOD1 mRNA knockdown	FDA approved 2023; ATLAS Ph3	Reduced SOD1/NfL; slowed ALSFRS-R	[26,27,28,31 37,38,40,41]

3.4 Muscular dystrophy (neuromuscular, not strictly neurodegenerative, but instructive)

Delandistrogene moxeparvovec (Elevidys), an AAVrh74-delivered microdystrophin construct for Duchenne muscular dystrophy, illustrates both the promise and the risks of systemic AAV gene therapy. The evidence for efficacy is mixed: the AAN Evidence in Focus review, which covered four studies with available data (two Class I and two Class III), noted that both Class I studies did not meet their primary functional endpoint (North Star Ambulatory Assessment), with small, non-significant secondary effects [17], and a meta-analysis of five trials (190 participants) found a modest effect on motor function at one year, with greater effects in participants aged 4 to 5 years [42]. Two-year outcomes from the phase 3 EMBARK trial in ambulatory patients [18] and 3- and 5-year data presented at scientific conferences [43] are extending the durability evidence, which is still accruing. Safety has since reshaped the label: after reports of fatal acute liver failure in non-ambulatory patients, the FDA in November 2025 added a Boxed Warning for serious liver injury and acute liver failure, limited the indication to ambulatory patients aged 4 years and older with a confirmed DMD mutation, and removed the non-ambulatory indication [19]. A pooled analysis of 236 trial participants (227 ambulatory and 9 non-ambulatory) reported treatment-related serious liver injury in 13 patients (5.5%) and myocarditis in 2 (0.8%) during the first 60 days, and two treatment-related deaths from acute liver failure in non-ambulatory patients [44].

4. Cross-Cutting Regulatory and Technical Themes

Regulatory acceleration for ultra-rare disease: The FDA's 2026 draft Plausible Mechanism Framework, which FDA leaders illustrated with the CPS1/KJ case, proposes evidentiary considerations for approval of individualized genetic therapies when large randomized trials are not feasible (an identified molecular abnormality, a therapy that targets it, well-characterized natural history, confirmation of target engagement or editing, and improvement in clinical outcomes, disease course, or established biomarkers), a potential structural shift aimed at diseases affecting single patients or handfuls of families [12,13,15,16].

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Because the guidance is a non-binding draft, its practical scope will depend on the final version.

AAV immunogenicity and redosing: Pre-existing anti-AAV antibodies are common and vary by serotype; in one healthy-population study, neutralizing factors were detected in about 59% of samples for AAV2 and 50% for AAV1, but in only 19% for AAV8 and 3% for AAV5 [45]. Such antibodies can constrain eligibility (for example, the post-marketing control group for Genglycos comprises patients who cannot be treated because of anti-AAV8 antibodies [3]) and can limit redosing of in vivo AAV therapies, a limitation that LNP-delivered mRNA/CRISPR platforms (as used in the CPS1 case) may partially circumvent [9].

Long-term follow-up burden: FDA guidance recommends long-term follow-up observations for gene therapy products, with the duration depending on vector type and risk, from about five years for non-integrating AAV vectors to 15 years for integrating vectors and genome-editing products [46]; sponsors of accelerated approvals may also take on post-marketing monitoring commitments of their own (10 years for Genglycos [3]). These requirements reflect uncertainty about the durability and the theoretical genotoxic or oncogenic risk of integrating or long-persisting vector genomes; disease-specific registries such as the World Federation of Hemophilia Gene Therapy Registry have emerged to manage follow-up at scale [47].

Manufacturing and scalability: Individualized and ultra-rare disease programs face a tension between conventional AAV vector manufacturing and the need for rapid, low-volume, patient-specific production, a problem the roughly six-month KJ-therapy timeline suggests may be solvable for LNP/RNA-based platforms even if not yet for viral vectors [9,10].

5. Future Directions and Challenges

Three trends are converging: (1) the first reported substantial slowing of clinical progression in a genuinely difficult neurodegenerative target (Huntington's disease, via AMT-130), which now awaits regulatory review and confirmation beyond an external-control comparison; (2) a working proof-of-concept for on-demand, single-patient gene editing, now accompanied by a draft regulatory framework intended to help scale it; and (3) accumulating real-world and long-term data on the first wave of AAV and ASO approvals that is beginning to clarify both durable benefit and previously undercharacterized risks (immunogenicity, rare severe adverse events such as acute liver failure, variable individual response). The central unresolved challenges, AAV immunogenicity, delivery to the CNS, manufacturing scalability for individualized therapies, and multi-year safety surveillance, remain the rate-limiting factors for broader translation across both rare and neurodegenerative disease.

6. Conclusion

Gene therapy for rare and neurodegenerative diseases has moved from proof-of-concept to regulatory approval, with new products, a first patient-specific editing therapy, and a draft framework for individualized therapies all emerging within roughly two years. Progress in Huntington's disease, Parkinson's disease, and ALS remains uneven and, in several cases, rests on preliminary evidence, while AAV immunogenicity, CNS delivery, manufacturing, and long-term safety remain open problems. Rigorous confirmatory trials and long-term follow-up will determine which of these advances translate into durable clinical benefit.

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